INCH-POUND

MIL-P-82823(OS)
29 December 1989
SUPERSEDING
WS 20645
8 November 1983
(See 6.8)

MILITARY SPECIFICATION

PRIMER, ADHESIVE, CHLORINATED RUBBER BASE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification defines one type of chlorinated rubber-base adhesive primer referred to herein as the primer (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation
(see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8040

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STANDARDS

FEDERAL

FED-STD-313	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129	Marking for Shipment and Storage

SPECIFICATIONS

MILITARY

MIL-T-81533	1,1,1 Trichloroethane (Methyl Chloroform), Inhibited, Vapor Degreasing

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part c this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199	Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 429	Rubber Property - Adhesion to Rigid Substrates (DoD adopted)
ASTM D 1084	Viscosity of Adhesives
ASTM D 1875	Density of Adhesives in Fluid Form (DoD adopted)

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Material.** The primer shall be a chlorinated rubber-base adhesive compound.

3.3 **Chemical, mechanical, and physical properties.** Chemical, mechanical, and physical properties of the primer shall conform to the requirements specified in table I.

3.4 **Safety.**

3.4.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

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3.4.2 **Material Safety Data Sheet (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.4).

TABLE I. Chemical, mechanical, and physical properties.

Property	Limits	
	Minimum	Maximum
Solids content, percent	22.0	27.0
Specific gravity, 25 degrees Celsius (°C)	0.92	0.97
Viscosity, centipoise	85	165
Peel strength, rubber-to-metal, pounds/inch of width 1/	90	---

1/ Minimum of 90 percent cohesive failure in the rubber is required.

3.5 **Shelf life.** The shelf life of the primer shall be 12 months after date of manufacture when warehouse stored at ambient temperature in the original sealed containers.

3.5.1 **Shelf life extension.** The shelf life of an individual lot may be extended for an additional 12-month period provided the material, when sampled and tested, conforms to the viscosity and solids content requirements.

3.6 **Workmanship.** The workmanship shall be such that the primer is uniform in appearance and free from visible contamination and defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall

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become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use (see 3.4.2).

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. First article and quality conformance inspections.

Examination and Tests	Requirement	Test Method
Visual examination	3.6 and section 5	4.5.1
Solids content test	3.3	4.5.2
Specific gravity test	3.3	4.5.3
Viscosity test	3.3	4.5.4
Peel strength test: rubber-to-metal	3.3	4.5.5

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4.4.1 **Sampling for inspection.**

4.4.1.1 **Lot.** For inspection purposes, a lot shall consist of all the adhesive primer offered for acceptance at one time from one of the following:

a. A single batch (see 6.5.1).

b. A continuous process (see 6.5.2).

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.4.1.2 **Selection of samples.** Each lot shall be randomly sampled in accordance with the following:

a. Supplier - A representative sample of sufficient size to perform the inspections specified in 4.5.1 through 4.5.5.

b. Contracting activity - The number of containers in accordance with table III for the inspections specified in 4.5.1 through 4.5.5.

4.4.1.3 **Sample material.** Sufficient material shall be selected at random from each sample container to perform all the inspections specified in 4.5.1 through 4.5.5.

TABLE III. Sampling for inspection.

Number of containers in the lot	Number of sample containers to be selected for inspection
1-3	1
4-40	2
41-110	3
111-180	4
181-300	5
301-500	7

4.4.1.4 **Noncompliance of samples.** Failure of one or more of the sample containers of material to meet all of the specified requirements shall be cause for rejection of the lot.

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4.5 **Methods of inspection.** The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Each sample of primer shall be examined visually for conformance to 3.6 and section 5.

4.5.2 **Solids content test.** The solids content of primer from each selected container shall be determined in accordance with the following:

a. Add 0.8 to 1.2 grams (g) of sample weighed to the nearest 0.1 milligram (mg) into a tared aluminum weighing dish.

b. Coat sides of weighing dish with sample.

c. Place dish with sample in a circulating-air oven at 154 ± 5 degrees Fahrenheit (^{0}F) for 2 hours ± 5 minutes.

d. Remove dish and contents from oven and place in desiccator to cool.

e. When cooled to room temperature, reweigh dish and contents to nearest 0.1 mg.

f. Calculate solids content as follows:

$$\text{Solids content, percent} = \frac{C - A}{B} \times 100$$

Where: A = weight of dish, g

B = weight of sample, g

C = weight of dish and contents after heating, g

g. Test two additional samples if the results of the first two determinations differ by more than 0.5 percent.

h. Record the average of two determinations to the nearest 0.1 percent.

4.5.3 **Specific gravity test.** The specific gravity of primer from each sampled container shall be determined in accordance with ASTM D 1875, except that the determination shall be made at 25 ± 1^{0}C and convert density to specific gravity. Record specific gravity to the nearest 0.01 unit.

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4.5.4 **Viscosity test.** The viscosity of primer from each sampled container shall be determined in accordance with ASTM D 1084, Method B, at 25 ± 1°C using a model LVF Brookfield viscometer or equivalent, with a number 2 spindle at 30 revolutions per minute. Prior to performance of the viscosity test, the test sample shall be thoroughly mixed until all solids are in suspension. The viscosity of the material from each sampled container shall be recorded to the nearest centipoise.

4.5.5 **Peel strength, rubber-to-metal test.** The peel strength, rubber-to-metal (1010 cold rolled steel), shall be performed on three test specimens from each selected container in accordance with ASTM D 429, Method B, with the following exceptions:

a. Clean bonding surfaces of steel strips by vapor degreasing with 1,1,1-trichloroethane conforming to MIL-T-81533. Gritblast with 60-mesh sand or 100- to 200-mesh zirconium silicate, then repeat vapor degreasing process.

b. A 1-inch by 1-inch bonding area of steel strips shall be primed with the primer being tested and dried 60 ± 5 minutes at 75 ± 5°F.

c. Standard test specimen shall consist of uncured rubber stock as described in 6.6, or equivalent.

d. Vulcanize assembled specimen at 500 ± 100 pounds per square inch (psi) and 320 ± 5°F for 32 ± 1 minutes. After vulcanizing, test specimen shall be cooled and maintained at room temperature for a minimum of 2 hours before testing. A fabric reinforcement may be added to elastomer tab to preclude tensile failure in tab.

e. Peel rate shall be 20 inches per minute (indicated).

f. Peel angle shall be 45 degrees.

g. Conduct peel test at 75 ± 5°F.

h. Record the average of three determinations to the nearest pound per inch of width.

4.6 **Inspection of packaging.** The primer containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The primer shall be packaged to afford adequate protection against loss or contamination the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.5.

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The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178 or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

d. Net weight of contents.

e. Lot number and date of quality conformance inspection.

f. Shelf life and inspection/test date for extension (see 3.5.1). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 12 MONTH PERIOD PROVIDED THE PRIMER, UPON RETEST, CONFORMS TO THE VISCOSITY AND SOLIDS CONTENT REQUIREMENTS SPECIFIED IN MIL-P-82823."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The primer specified herein is intended for use in bonding Standard Missile Mark 104 Dual Thrust Rocket Motor components.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

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c. Whether first article is required (see 3.1 and 6.3).

d. Material Safety Data Sheets (see 3.4.1 and 6.4).

e. Packaging required, if other than as specified in 5.1.1.

f. Special marking, if other than as specified in 5.2.2.

6.3 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the primer sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 **Material Safety Data Sheet (MSDS).** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed MDSDs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 **Definitions.**

6.5.1 **Batch.** A batch is that amount of adhesive primer made as one unit in the manufacturing process.

6.5.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the adhesive primer is accumulated for a lot.

6.6 **Typical materials.** [1/] Rubber Stock HC-600 (an uncured rubber stock) manufactured by Lord Corporation is typical of the product used to perform the peel adhesion tests. Chemlok 205 manufactured by Lord Corporation is typical of the material described herein.

[1/] This listing is not intended to indicate that only specific materials meets the requirements specified herein.

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6.7 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.8 Supersedure information. MIL-P-82823 incorporates the following engineering change proposals (ECPs) and specification change notices (SCNs):

ECPs	SCNs
----	SCN 1 (12/10/86)
MTA026 (10/15/87)	SCN 2 (8/17/87)
MTA046 (1/19/88)	

Preparing Activity:
NAVY-OS
(Project 8040-N150)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-82823	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
PRIMER, ADHESIVE, CHLORINATED RUBBER BASE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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